

NETWORK TOPOLOGIES

TASK – 1 Difference between star, mesh and bus topologies.

- Star Topology:
 - Central switch/hub connected directly to 5 individual nodes in a radial layout.
 - Viewers connect individually to central distribution servers. If one user experiences a network drop, it only affects that individual without disrupting the live stream for millions of other viewers.
- Bus Topology:
 - Central backbone cable terminated at both ends with drop-line taps connecting nodes.
 - Multiple display panels and audio modules connect directly to a single shared cable bus across the vehicle, minimizing wiring costs and keeping installation simple.
- Mesh Topology:
 - Diagram is fully interconnected mesh where every node has a direct connection to every other node.
 - Direct peer-to-peer data paths are created between callers. If one routing path becomes congested, data packets are dynamically rerouted through an alternate path without dropping the call.

TASK – 2 Comparsion table of topologies.

TOPOLOGIES	PROS	CONS
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Star topologies	1. Centralized network control & monitoring. 2. Single node failure doesn't affect others. 3. Easy to add/remove devices on the fly. 4. Simple troubleshooting and cable layout.	1. Central switch/hub failure crashes entire network. 2. Requires more cabling than bus topology. 3. Switch port count caps maximum connections. 4. High-grade central hardware can be costly.
Mesh topologies	1. Highest fault tolerance via redundant paths. 2. Dedicated lines eliminate bandwidth bottlenecks. 3. Enhanced security via direct point-to-point links. 4. Adding nodes does not interrupt current connections.	1. Extremely expensive due to extensive cabling. 2. Highly complex setup requiring multiple network ports. 3. Physical cable management gets messy quickly. 4. High power and hardware resource requirements.

Bus topologies	1. Minimal cabling needed, very budget-friendly. 2. Very simple to set up in linear layouts. 3. Ideal for small, low-traffic environments. 4. No active central network hardware required.	1. Backbone cable break brings down entire network. 2. Frequent data collisions as device count increases. 3. Difficult to isolate specific cable faults. 4. Network speed drops sharply under heavy load.
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TASK – 3 Choosing the perfect topologies for gaming .

- For a game like PUBG or Free Fire I would pick a Hybrid Star-Mesh setup. In this setup mobile players connect to edge servers that are arranged in a Star pattern. Meanwhile the servers that handle the games functions are connected in a Full Mesh pattern. This setup helps players get latency because they connect to servers that are close, to them. It also makes the system more reliable because if one server stops working the mesh part of the system sends the data through a path.. It keeps the costs under control because the expensive part of the mesh is only used for the most important servers.

TASK – 4 Connection of computer.

- I recommend using a Star topology with an eight-port central Ethernet switch to link all the warehouse workstations. This Star topology keeps setup costs and makes troubleshooting simple. If one workstation cable breaks the other seven order-processing stations stay online. Expanding later is also easy with a Star topology— plug an Ethernet cable into an open switch port, for a new computer.